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1. This problem concerns the permutation $\mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 1 & 6 & 7 & 2 & 4 & 3 \end{pmatrix} \in S_7$

(a) Write μ as a product of disjoint cycles.

$$\mu = (1 \ 5 \ 2)(3 \ 6 \ 4 \ 7)$$

(b) Is μ even or odd?

$$\mu = \underbrace{(1 \ 5 \ 2)}_{\text{even}} \cdot \underbrace{(3 \ 6 \ 4 \ 7)}_{\text{odd}} = \boxed{\text{odd}}$$

(c) Find the order of μ .

$$\gcd(3, 4) = \boxed{12}$$

(d) Find μ^{1000} . From above we know $\mu^{12} = \hat{i}$.

$$\text{So } \mu^{1000} = \mu^{12 \cdot 83 + 4} = \mu^{12 \cdot 83} \mu^4$$

$$= (\mu^{12})^{83} \mu^4 = \hat{i}^{83} \mu^4 = \mu^4 = (1 \ 5 \ 2)^4 (3 \ 6 \ 4 \ 7)^4$$

$$= (1 \ 5 \ 2)^3 (1 \ 5 \ 2) = \boxed{(1 \ 5 \ 2)}$$

$$\begin{array}{r} 83 \\ 12 \overline{) 1000} \\ \underline{96} \\ 40 \\ \underline{36} \\ 4 \end{array}$$

(remainder)

2. Write the permutation $(215)(3142)^{-1}$ as a product of disjoint cycles.

$$(2 \ 1 \ 5)(3 \ 1 \ 4 \ 2)^{-1} = (2 \ 1 \ 5)(2 \ 4 \ 1 \ 3)$$

$$= \boxed{(2 \ 4 \ 5)(3 \ 1)}$$

3. Prove that if a permutation σ is even, then σ^{-1} is also even.

Proof (Direct) Suppose σ is even.
This means σ can be written as
a product of an even number $2k$
of transpositions, as

$$\sigma = (a_1 b_1)(a_2 b_2)(a_3 b_3) \cdots (a_{2k-1} b_{2k-1})(a_{2k} b_{2k})$$

for some $k \in \mathbb{N}$. Then

$$\begin{aligned}\sigma^{-1} &= (a_{2k} b_{2k})^{-1}(a_{2k-1} b_{2k-1})^{-1} \cdots (a_3 b_3)^{-1}(a_2 b_2)^{-1}(a_1 b_1)^{-1} \\ &= (a_{2k} b_{2k})(a_{2k-1} b_{2k-1}) \cdots (a_3 b_3)(a_2 b_2)(a_1 b_1)\end{aligned}$$

Thus σ^{-1} is a product of an even
number $2k$ of transpositions, so
 σ^{-1} is even. 